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Spatiotemporal Analysis of Traffic Patterns on Malioboro Street Using Mobile Positioning Data

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***Abstract*—Traffic congestion continues to pose major challenges in Indonesian cities, particularly in tourist hubs like Yogyakarta, where streets such as Malioboro are frequently congested due to high vehicle volumes. This study investigates the application of Active Mobile Positioning Data (MPD) to analyze traffic speed and flow patterns on Malioboro Street, Yogyakarta. The research involved extensive preprocessing of 2.2 million raw GPS data points, reducing them to 229,826 quality-filtered entries through several stages, such as removing stay locations, map matching using KDTree, and directional alignment. A total of 2,316 user travel trajectories were reconstructed using Hidden Markov Models (HMM) and Viterbi algorithms. The results show that weekdays exhibited higher average speeds (15.54 km/h) and more regular traffic flow patterns, particularly during morning hours, whereas weekends showed slower and more variable traffic behavior due to tourist activity. Validation against Google Maps data and official traffic surveys using Dynamic Time Warping (DTW) and Pearson's correlation confirmed a moderate to strong agreement, particularly for weekday flow (r = 0.79) and weekend speed (r = 0.63). This study demonstrates that active MPD can effectively capture urban traffic dynamics and support traffic policy planning.**

***Index Terms*—mobile positioning data, GPS, traffic speed, traffic flow, path reconstruction**

I. Introduction

Traffic congestion has become one of the major challenges facing many cities in Indonesia. Several factors contribute to the traffic problem, including rapid population growth, increased car ownership, and inadequate road infrastructure. Traffic congestion leads to delays, longer lines, and even economic losses while also making travel less comfortable and causing more vehicle emissions that harm the air [1].

Yogyakarta, one of Indonesia's most popular tourist destination cities, experiences particularly high traffic volumes in its urban center. Malioboro Street, the city's most iconic and busiest street, frequently faces traffic congestion due to its role as a central hub for tourism, commerce, and local events. Understanding how traffic flows in such high-demand areas is vital for developing better urban mobility strategies.

To address complex traffic issues like those in Yogyakarta, it is important to analyze how fast and how many vehicles are moving. Currently, Indonesia collects traffic data using manual and mechanical methods, following the guidelines set by the Directorate General of Highways (*Direktorat Jenderal Bina Marga*), Ministry of Public Works and Housing of the Republic of Indonesia [2]. However, manual methods are both expensive and take a lot of time. Mechanical surveys, on the other hand, need a large amount of expensive equipment [3]. With advancements in digital technology, positioning data obtained from smartphones has emerged as a promising alternative. This data offers spatiotemporal information that reveals human movement patterns [4]. Among the available types, active Mobile Positioning Data (MPD) offers higher accuracy but requires user consent and specific data collection frameworks [5].

Many studies have demonstrated that MPD can be useful for analyzing traffic. Forghani et al. [4] underscored its ability to accurately reconstruct travel routes, while Wang et al. [6] leveraged trajectory data to characterize urban congestion. Meanwhile, Zheng et al. [7] enhanced trajectory-based analyses through time interpolation methodologies. Building on these foundational studies, the present research examines how active MPD can be harnessed to reveal traffic patterns in Yogyakarta, offering fresh insights into mobility dynamics in a tourism-driven urban context.

The contributions of this work are to: (1) identify traffic speed and flow patterns in tourism-driven urban cities using active MPD, and (2) evaluate these patterns using relevant datasets, including Google Maps and official government surveys. This work addresses the limited use of active MPD in Indonesia and enhances understanding of urban traffic dynamics through data-driven analysis. The remainder of this manuscript is structured as follows: Section II reviews the relevant literature; Section III details the research methodology; Section IV presents and discusses the results; and Section V offers the conclusions.

II. Related Works

Mobile Positioning Data (MPD) has been used to gain insights into people mobility in many fields. During the COVID-19 pandemic, MPD used to analyze citizen movement and track virus spread. In Indonesia, Widyawan et al. [8] leveraged raw MPD in collaboration with a national telecommunications provider to map mobility trends at the provincial

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level. Their findings showed a notable shift: while average travel distances dropped significantly due to restrictions, the number of short trips actually increased. To address this, Khayati et al. proposed an academic sandbox system to enable secure and ethical access to MPD [9]. Leveraging the ELK stack (Elasticsearch, Logstash, Kibana), the system offers interactive dashboards—overview, origin-destination, and urban hotspot—while enforcing role-based access and anonymization to safeguard sensitive data. Their preprocessing pipeline includes data cleaning, clustering, oscillation filtering, and reverse geocoding, enabling privacy-preserving yet detailed mobility analytics. This framework was effective in analyzing movement trends in Yogyakarta during the pandemic.

In the domain of traffic analysis, active MPD combined with GPS data has been employed to detect speed and flow dynamics. Wang et al. [6] conducted a large-scale study on urban congestion in Beijing using GPS data from over 28,000 taxis. They developed a real-time interactive system using ST-matching for map alignment and visualized traffic flow using a pixel map based on more than 379 million sample points collected over 24 days.

Rakhman et al. [10] focused on reconstructing travel routes from low-frequency GPS datasets similar to active MPD data. Their approach included preprocessing raw GPS data, map matching using a KD-Tree algorithm, and path reconstruction via the A* shortest path algorithm. They successfully reduced data volume with immobility filters but lacked trajectory direction differentiation and detailed preprocessing methods.

Tramontana et al. [11] analyzed GPS data to identify flow and flow density on road segments, employing trajectory cleaning, map matching, and calculations of user counts and average speeds. Their method effectively identified high-traffic flows and speeds, highlighting congestion areas.

Satrinia et al. [12] predicted traffic speeds in Bandung using Support Vector Regression (SVR) on taxi GPS data. Their approach included preprocessing GPS data, map matching with Dijkstra's algorithm, and leveraging road map data for accuracy in predicting future traffic speeds.

This study focuses on Malioboro Street, Yogyakarta, using active MPD to identify traffic speed and flow patterns. Unlike Wang et al., it emphasizes traffic pattern identification and visualization rather than developing an interactive system. Similar to Rakhman et al., the study addresses preprocessing, map matching, and route estimation from GPS data, tailored for low sampling frequency and non-stationary data. It integrates techniques from Wang et al. and Rakhman et al., with modifications for enhanced accuracy and efficiency. Validation against ground truth data, akin to Wang et al., ensures the reliability of findings.

III. Methodology

The process began with data preprocessing to remove noise and outliers, followed by path reconstruction to generate travel routes, and subsequently, the calculation of traffic speed and flow. The resulting analyses were then validated using Google Maps data and survey data from the Yogyakarta Transportation Agency (*Dinas Perhubungan DIY*). The validation process was performed using Dynamic Time Warping (DTW) [13] and Pearson's correlation coefficient [14] to measure alignment and correlation between each data point.

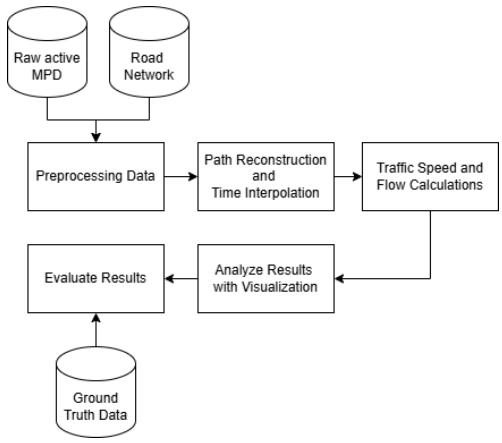

Fig. 1. Methodology Process Flow

A. Data Acquisition

The active MPD dataset was obtained from one of Indonesia's largest telecommunications companies. The data were captured from user smartphones via satellite-based positioning (GPS) with assisted-GPS (A-GPS) to improve urban accuracy, achieving typical positional accuracy of 5–10 meters under optimal conditions and 10–30 meters in dense urban areas, with relatively low temporal granularity due to wide intervals between successive records. This study utilizes data from January to February 2022, containing unique user identifiers, latitude–longitude coordinates, and timestamps.

Shapefiles of road networks in Yogyakarta were obtained from OpenStreetMap (OSM), comprising 896 road segments, as depicted in Table I. Each feature in this shapefile includes detailed geometrical data of the road. The road segments were used to map the city's road network to the GPS data, which was not provided in the active MPD dataset.

TABLE I
Shapefile Data of Road Segments

name	geometry
Jalan Taman	LINESTRING (110.36943 -7.80153, 110.36945 -7.8)
Jalan Tentara	LINESTRING (110.35918 -7.78753, 110.35932 -7.7)
Jalan Ratu	LINESTRING (110.37631 -7.78300, 110.37633 -7.7)
Jalan Kartini	LINESTRING (110.37685 -7.78098, 110.37680 -7.7)
Jalan Suroto	LINESTRING (110.37462 -7.78299, 110.37402 -7.7)

To validate the proposed method, two types of ground truth data were employed. The speed ground truth data were obtained from Google Maps historical traffic speed information, accessed via OutscraperAPI. This dataset provides average hourly speed data along Malioboro Street, extracted from anonymized aggregated user location data collected by Google. The data were spatially aggregated at the road

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segment level as defined by Google Maps, with an hourly temporal resolution.

For validating traffic flow, survey data from Yogyakarta Transportation Agency conducted in 2022 were used. These data include traffic volumes in standardized passenger car units per hour (pcu/hr), measured manually at several fixed observation points along Malioboro Street during different time periods throughout the day. The spatial resolution corresponds to specific survey locations along the street, while the temporal resolution varies by survey period, typically aggregated in hourly intervals. These datasets serve as reliable references for assessing the performance of the MPD-based estimation.

B. Preprocessing Data

Before analysis, the MPD data underwent several preprocessing steps as follows:

1) Preprocessing Data: Prepares data by converting UNIX timestamps to local time, filtering points beyond a 1.5km radius from the targeted area, removing duplicates to avoid bias, and sorting data chronologically.
2) Eliminating Stay Locations: This step removes stationary points like home or office locations using MovingPandas StopSplitter() function based on a 200-meter distance and 30-minute duration threshold.
3) Eliminating Outliers Distant from the Road Network: Points located more than 20 meters away from the nearest road segment are excluded using the Shapely buffer function, in order to retain only points associated with on-road movement.
4) Map Matching with KDTree Algorithm: Map matching with KDTree corrects GPS points to the nearest road segment. This process ensures that the GPS data aligns with the road network.
5) Adjusting Points to Road Direction: GPS points are adjusted to align with the correct road direction using a MultiDiGraph representation of the road network, ensuring accuracy in user movement.
6) Trajectory Segmentation based on Observation Time: Data is segmented into trajectories based on 10-minute time gaps between GPS points using MovingPandas ObservationGapSplitter() function.

C. Path Reconstruction

After preprocessing, travel routes were estimated using PyTrack library, a Python toolbox designed for vehicle trajectory reconstruction [15], which uses Hidden Markov Model (HMM) and Viterbi algorithm for map matching [16] [17].

In the HMM formulation used by PyTrack, the noisy GPS measurements are the observations, while the candidate road points that are spatially close to each GPS point represent the hidden states. PyTrack first prepares the road network by interpolating it at a step distance of 5 meters for more accurate map matching results. It then uses the get_candidates method to generate a dictionary of candidate points, searching within a 30-meter radius around each GPS point.

The emission probabilities were computed from the perpendicular distance between each GPS point and its candidate road segments, with shorter distances producing higher probabilities. Transition probabilities were calculated from shortest-path distances and estimated travel directions between candidates along the road network. Then, a directed acyclic trellis graph was constructed, where each node represents a candidate and each edge represents possible transitions between two candidates. The Viterbi algorithm was used on this trellis to find the most probable sequence of candidates. The path with the highest cumulative joint probability of emission and transition probabilities represents the most probable sequence of road segments traversed by the user.

Following the route reconstruction, time interpolation was performed to fill missing time values at reconstructed route points using a linear method, that is, averaging the time of surrounding points to ensure complete and consistent time information for the user's travel path. The linear interpolation was calculated using Equation 1, where Δ_t represents the time difference between the two observed GPS points with timestamps t_1 and t_2. The missing time values are estimated by dividing Δ_t by $n+1$ (Equation 2), where n is the number of new points without timestamps.

$$\Delta t = t_2 - t_1 \tag{1}$$

$$\delta_t = \frac{\Delta_t}{n+1} \tag{2}$$

$$t_i = t_i + (i \times \delta t) \tag{3}$$

Each interpolated point timestamp t_i was then estimated using Equation 3, where i represents the index of the new point. This process continues until all data rows are examined, resulting in complete route data with consistent timestamps, accurately representing the temporal movement of an individual based on the reconstructed travel paths.

D. Speed and Flow Calculation

After obtaining the reconstructed trajectories from the previous steps, the speed and flow on Malioboro Street were calculated. The input for this calculation is a DataFrame containing the reconstructed routes and their corresponding timestamps. Initially, only the trajectory points located within the Malioboro Street polygon are selected using a shapefile geometry, which ensures the analysis is focused exclusively on movements within the specified street.

$$v_j = \frac{Dist(p_j, p_{j+m})}{(t_{j+m} - t_j)} \tag{4}$$

$$v_{\text{avg}} = \frac{\sum_j^m v_j}{m} \tag{5}$$

The average speed is calculated from the trajectories passing through Malioboro Street. The speed for each segment of a trajectory, denoted as v_j, is calculated by dividing the distance between two consecutive GPS points, p_j and p_{j+m}, by the time difference, $t_{j+m} - t_j$. The overall average speed (v_{avg}) was

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then computed as the sum of all individual trajectory speeds (v_j) divided by the total number of trajectories (m). On the other hand, the flow is calculated by counting the number of trajectories that pass through Malioboro Street during each hour. The counting was performed on the trajectories that have already been filtered to include only points located within the Malioboro Street polygon.

IV. Results and Discussion

A. Preprocessing Data

Table II summarizes the results of the active MPD data preprocessing. From the initial 2,195,131 GPS points, which were reduced to 229,826 GPS points for path reconstruction input. The preprocessing steps successfully eliminated non-relevant points, enhancing data quality and accuracy for further analysis.

TABLE II
Results of Active MPD Data Preprocessing

No	Method	GPS Points	Percentage
1	Preprocessing Data	2,195,131	100%
2	Eliminating Stay Locations	342,123	15.58%
3	Eliminating Points distances above 20m from Road Network	340,744	15.52%
4	Map Matching with KDTree Algorithm	340,744	15.52%
5	Adjusting Points with Road Direction	229,826	10.46%

As seen in the table, the most significant preprocessing step is eliminating stay locations. This process eliminated 85% of the original data, meaning that most of the data were obtained when users were stationary, possibly due to activities such as being at home, work, or other static locations.

B. Path Reconstruction

The path reconstruction process estimates user travel routes on Malioboro Street, as illustrated in Figure 2. With around 2,316 trajectories reconstructed, the path reconstruction process successfully estimates all of the user travel routes. After this process, each trajectory was refined to follow the road network, aligning with the identified path segments. This refinement ensures that the reconstructed routes reflect real travel behavior.

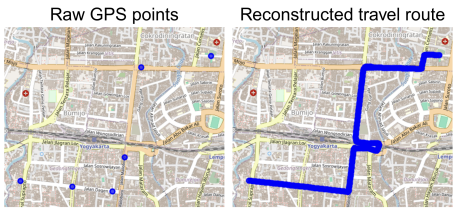

Fig. 2. Sample of Travel Route Estimation Results

After path reconstruction, the time interpolation process fills missing time values at reconstructed route points, ensuring complete and consistent time information for the user's travel path.

C. Speed Calculation Results

Speeds are then calculated from the reconstructed travel routes, as shown in Table III. The speed calculation results in a DataFrame with features including the unique user ID, start and end times, distance traveled, duration, speed, and bearing. The speed DataFrame provides valuable insights into traffic speed patterns on Malioboro Street, enabling further analysis of speed and flow patterns.

TABLE III
Features of Speed DataFrame

Feature	Description	Data Type
maid_trajectory	Unique ID representing each user journey	object
start_t	Start time of GPS point observation	datetime64
end_t	End time of GPS point observation	datetime64
distance	Distance traveled from start GPS point to end GPS point in meters	float64
duration	Duration of travel from start GPS point to end GPS point in seconds	float64
speed	Speed of the journey calculated as distance per duration in km/h	float64
bearing	Direction of movement of the GPS point in degrees	float64

D. Speed and Flow Pattern Analysis

The speed and flow patterns on Malioboro Street were analyzed based on the reconstructed travel routes. As shown in Figure 3, the heatmap analysis showed that the average traffic speed on Malioboro Street was higher on weekdays (15.54 km/h) compared to weekends (12.88 km/h). On weekdays, the highest speeds were observed in the morning (6-8 AM) with peaks up to 28 km/h, indicating that traffic on Malioboro Street is less affected by typical morning rush hours. Speeds gradually declined throughout the day, reaching their lowest in the afternoon (3-6 PM) at 9.7 km/h, likely due to increased traffic as people returned home. The speed increased again at night, attributed to the Car Free Night (CFN) policy from 6-9 PM. On weekends, the speed pattern varied more, with the highest speeds also in the morning but a sharp decline by afternoon, reaching as low as 6 km/h. The evening speeds were unpredictable due to higher tourist and recreational activities.

The traffic flow analysis on Malioboro Street, based on speed data from hourly trajectories, reveals distinct patterns shown in Figure 5. On weekdays, traffic increases gradually from 6 AM, peaks at around 3 PM, and drops sharply by 6 PM, likely due to the CFN policy. Traffic flow is consistent on weekdays with a lower standard deviation of 3.75, reflecting regular commuting patterns. On weekends, traffic flow is generally lower between 9 AM and 5 PM, with a higher standard deviation of 4.97, indicating greater variability. This variability is most pronounced in the afternoon, likely due to varying weekend activities such as recreation and events.

The analysis also identified several anomalies where traffic speeds were much higher or lower than the general trend. These outliers might have been caused by special events, changes in the road, weather conditions, driver choices, or

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TABLE IV
DATAFRAME RESULTS OF SPEED CALCULATION

	maid_trajectory	start_t	end_t	distance	duration	speed	bearing
0	0lb75026-0e33-aa15-b4ca-65a7afe9c6dd_1	2022-01-27 14:26:27	2022-01-27 14:28:17	159.10	110.0	5.20	187.5
1	028de778-c3b6-4993-9f4c-339b33be244b_12	2022-01-23 21:10:28	2022-01-23 21:13:34	1144.8	186.0	22.1	187.2
2	028de778-c3b6-4993-9f4c-339b33be244b_22	2022-01-30 06:53:13	2022-01-30 06:53:27	75.66	14.0	19.4	188.6
3	037f297S-7S6S-4ce6-8860-8db2228c3cdf_8	2022-01-31 01:24:43	2022-01-31 01:28:42	1144.8	239.0	17.2	187.2
4	0419eb09-110d-435e-9b9fLa697ab3567fc_10	2022-01-24 06:27:58	2022-01-24 06:29:09	323.08	71.0	16.3	186.8

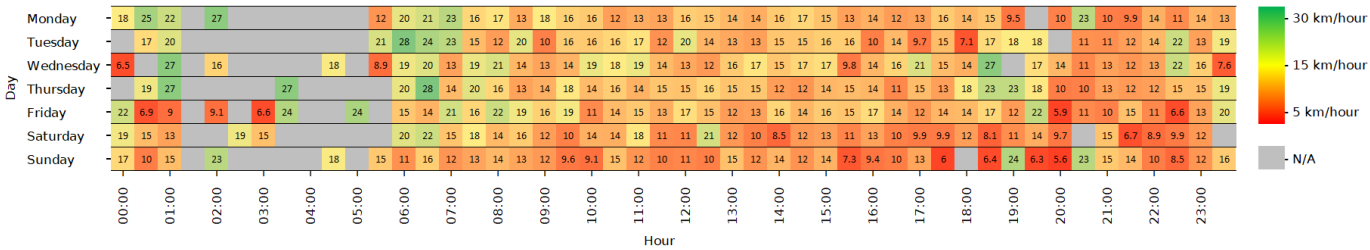

Fig. 3. Heatmap of Average Speed Distribution by Day and Hour

other unexpected reasons. Such deviations can significantly skew average speed calculations and may obscure typical traffic behavior if not properly accounted for. However, they also present valuable opportunities to uncover context-specific anomalies, such as road closures during festivals or sudden congestion due to accidents. Spotting these irregular patterns shows why more research and extra data collection are needed to better understand what causes these speed changes and to ensure the results are not too generalized.

E. Evaluation with Ground Truth Data

The speed and flow results from active MPD were checked against ground truth data from Google Maps and surveys by the Yogyakarta Transportation Agency. Using Dynamic Time Warping (DTW) and Pearson's correlation coefficient, the study measured how closely MPD data matched the real-world records.

TABLE V
DTW DISTANCES RESULTS FOR SPEED COMPARISON

Day	DTW Distances
Monday	37.37
Tuesday	37.94
Wednesday	65.48
Thursday	37.97
Friday	56.13
Saturday	33.36
Sunday	40.85

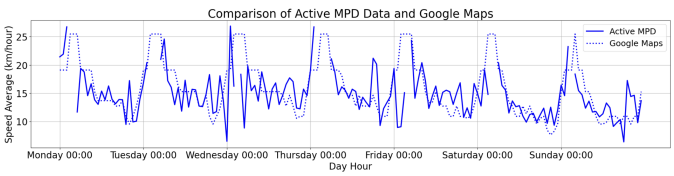

Fig. 4. Line Chart of Speed Comparison between Active MPD and Google Maps Data

As seen in Table V, the results indicate that DTW distances vary across different days of the week, with Saturdays showing the best match at a lowest distance of 33.36 and Wednesdays the worst at the highest distance of 65.48. Weekends generally have lower DTW distances compared to weekdays, suggesting that active MPD speed data aligns better with Google Maps data on weekends. A line chart visualization in Figure 4 further clarifies this, revealing significant differences on days with high DTW distances, like Wednesdays and Fridays, where MPD data shows greater fluctuations than Google Maps data. Conversely, days with low DTW distances, such as Saturdays and Mondays, exhibit more similar patterns, indicating better alignment.

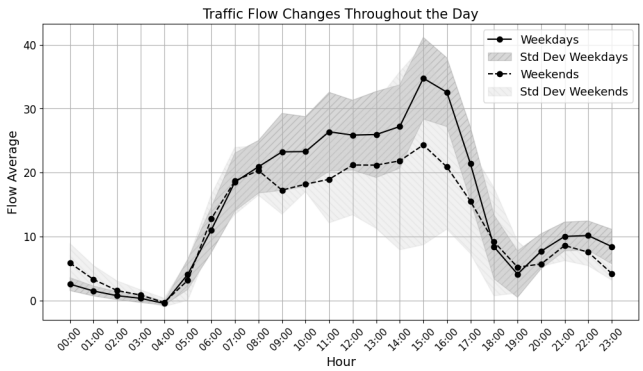

Fig. 5. Line Plot of Traffic Flow Changes Based on Weekdays and Weekends

Pearson correlation coefficient confirms a better linear relationship on weekends, with an overall correlation of 0.52, reflecting a moderate agreement between the datasets. Despite providing a reasonable estimate of general speed trends, active MPD data displays variability and some inaccuracies.

Similarly, traffic flow comparisons show stronger alignment during weekdays (Pearson's correlation coefficient of 0.79) between active MPD and ground truth data. However, weekend

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comparisons exhibit greater variability (Pearson's correlation coefficient of 0.44), suggesting less consistent traffic flow patterns during weekends.

Overall, while active MPD data offers valuable insight into speed and traffic flow trends, improving data collection and processing methods could improve accuracy, particularly to capture day-to-day variations and weekend dynamics. However, because this study focuses specifically on Malioboro Street, the generalizability of the results to other urban environments should be approached with caution. Streets with different functions, traffic regulations, or land use patterns (e.g., industrial zones, residential neighborhoods, or high-speed arterials) may exhibit distinct mobility behaviors that require tailored analysis and validation. Nevertheless, the approach is transferable to other urban settings by adjusting preprocessing thresholds, candidate search parameters, and validation datasets to suit local traffic conditions and GPS characteristics. In other tourism districts, similar weekday–weekend patterns in speed and flow could be observed, while central business areas may show more pronounced peak hour congestion during weekdays and reduced weekend variability. Cities with different structures in the street network may also require additional calibration of the map matching process. Thus, while adaptable, the results should be interpreted within the context of the specific urban environment.

Furthermore, the inherent limitations of active MPD data must be acknowledged. These include demographic biases, as data is often collected from smartphone users with certain socioeconomic characteristics, potentially underrepresenting groups such as older people or those without constant data connectivity. Low-frequency sampling may lead to data sparsity, affecting the accuracy of trajectory reconstruction, especially for short trips or brief traffic events. Network coverage variations can also lead to spatial inconsistencies. Collectively, these factors may influence the representativeness of the dataset and introduce uncertainties in the interpretation of traffic patterns. Addressing these limitations in future research will be crucial for developing more inclusive and reliable urban mobility models.

V. Conclusion & Future Work

This study conducted a comprehensive analysis of GPS time series data from active MPD to discern speed and traffic flow patterns on Malioboro Street, the most iconic tourist place in Yogyakarta, Indonesia. Through path reconstruction, vehicle speed and road density estimation, and comparison with ground truth data, the research achieved its objectives and derived significant insights.

The path reconstruction process helped estimate user travel routes, resulting in 2,316 trajectories ready for analysis. The findings showed apparent differences between weekdays and weekends: weekdays generally had higher average speeds and more predictable peak times, while weekends showed lower speeds and more varied traffic flows. Evaluation using Dynamic Time Warping (DTW) and Pearson's correlation showed that the best match between MPD and ground truth data happened on Saturdays, while Wednesdays had the most significant differences. In general, speed data from MPD matched better with ground truth on weekends, while traffic flow data aligned better on weekdays. Even though active MPD data gave a reasonably accurate view of traffic conditions, the accuracy still changed depending on the day and time, suggesting that more work is needed to fully understand the causes of these variations.

Future studies should focus on improving the accuracy of route estimation before analyzing the data. Expanding the data collection to include other factors, such as weather conditions, special events, and a more extended observation period, could also provide a deeper understanding of traffic patterns on Malioboro Street. In addition, utilizing more detailed ground truth data would further enhance the evaluation process and improve the quality and usefulness of urban mobility research.

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